🧪 Lab3 — MIME File Transfer over Router-on-a-Stick with Wireshark Analysis

A Cisco Networking Academy–style lab that blends classical networking with application-layer engineering. You will configure inter-VLAN routing, deploy a MIME-aware socket server, capture packets, and analyze how meaning flows through the stack.

📘 Lab Overview

Scenario

A client in one VLAN transfers files to a server in another VLAN using a custom MIME-based protocol (JSON header + binary payload). You will verify connectivity, observe encapsulation, and analyze packets using Wireshark.

This lab reinforces how Layer 2 segmentation, Layer 3 routing, TCP reliability, and application metadata interact.

🎯 Objectives

After completing this lab, you will be able to:

- Configure Router-on-a-Stick inter-VLAN routing
- Configure VLANs and trunking on a switch
- Deploy and run a MIME socket server and client
- Capture traffic using Wireshark
- Identify ARP, VLAN tags, TCP handshake, and application data
- Decode JSON headers in packet payload
- Troubleshoot network and application failures
- Document protocol behavior across OSI layers

🧰 Required Resources

- 1 Router (e.g., Cisco Systems ISR/virtual router)
- 1 Switch (Layer 2)
- 2 PCs or VMs
- Ethernet cables or virtual links

- Python 3 installed
- MIME file transfer project
- Wireshark installed (from Wireshark Foundation)

🗺️ Topology

```
None
PC1 (VLAN 10) ----\
                    \ access
                     \
                     [S1]---- trunk ----[R1]
                     /
                    / access
PC2 Server (VLAN 20)
```

📋 Addressing Scheme

Device	Interface	IP	Gateway
R1	G0/0.10	192.168.10.1/24	—
R1	G0/0.20	192.168.20.1/24	—
PC1-Client	NIC	192.168.10.10	192.168.10.1
PC2-Server	NIC	192.168.20.20	192.168.20.1

🧠 Background Theory

Router-on-a-Stick

A single physical interface carries multiple VLANs via 802.1Q tagging. Subinterfaces perform routing between broadcast domains.

MIME Metadata

MIME labels content type (e.g., image/png). Your protocol wraps files with JSON describing:

- MIME type
- Size

Why This Matters

Real systems — APIs, storage services, message brokers — depend on this exact pattern: structured metadata + binary payload over reliable transport.

🔧 Part 1 — Configure Network Devices

Step 1 — Router Configuration

```None
enable
configure terminal
hostname R1

interface g0/0
no shutdown

interface g0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0

interface g0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0

end
write memory
```

Physical Config CLI Attributes

```
If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO2901/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!



Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface g0/0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#interface g0/0.10
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip address 192.168.10.1 255.255.255.0
R1(config-subif)#interface g0/0.20
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)#ip address 192.168.20.1 255.255.255.0
R1(config-subif)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#write memory
Building configuration...
[OK]
R1#
R1#
```

Step 2 — Switch Configuration

```
None
enable
configure terminal
hostname S1

vlan 10
name CLIENTS

vlan 20
name SERVERS
```

```
interface g0/1
switchport mode trunk

interface fa0/1
switchport mode access
switchport access vlan 10

interface fa0/2
switchport mode access
switchport access vlan 20

end
write memory
```

S1

Physical Config CLI Attributes

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up


Switch>
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#vlan 10
S1(config-vlan)#name CLIENTS
S1(config-vlan)#vlan 20
S1(config-vlan)#name SERVERS
S1(config-vlan)#interface g0/1
S1(config-if)#switchport mode trunk

S1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

S1(config-if)#interface fa0/1
S1(config-if)#switchport mode access
S1(config-if)#switchport access vlan 10
S1(config-if)#interface fa0/2
S1(config-if)#switchport mode access
S1(config-if)#switchport access vlan 20
S1(config-if)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#write memory
Building configuration...
[OK]
S1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual down                  down
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down
 --More--
```

Step 3 — Configure Hosts

Set static IP addresses per table.

🔍 Part 2 — Verify Connectivity

From PC1:

```
None
ping 192.168.20.20
```

Verify:

PC1 Client

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.20

Pinging 192.168.20.20 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.20: bytes=32 time<1ms TTL=127
Reply from 192.168.20.20: bytes=32 time<1ms TTL=127
Reply from 192.168.20.20: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.20:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

- Successful replies
- ARP entries

Router checks:

```
None
show ip route
show ip interface brief
```

```
R1#
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0.20

R1#
```

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

Switch checks:

None

```
show interfaces trunk
show mac address-table
```

```
Vlan1                  unassigned      YES manual administratively dow
S1#
S1#
S1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Gig0/1      1,10,20

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      1,10,20

S1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    000a.f365.7a01    DYNAMIC     Gig0/1
  10    000a.f365.7a01    DYNAMIC     Gig0/1
  20    000a.f365.7a01    DYNAMIC     Gig0/1
S1#
```

🧪 Part 3 — Deploy MIME Server

On Server (PC2)

```
None
cd automation
python server/main_enhanced.py --host 0.0.0.0 --port 65432 --verbose
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                    python ⚠ + ⌄ ▯ 🗑 ··· | ⛶ ×

PS C:\Users\tharat\OneDrive\เดสก์ท็อป\NetWork\week01-mime-typing> pypython server/main_enhanced.
py --host 0.0.0.0 --port 65432 --verbose
[23:51:22] INFO: Starting server with host=0.0.0.0, port=65432, storage=storage
[23:51:22] INFO: Server listening on 0.0.0.0:65432
```

On Client (PC1)

```
None
cd automation
python client/main_enhanced.py assets/sample.png --host 192.168.20.20 --port
65432
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                    + ⌄ ··· | ⛶ ×

                                                                        python ⚠
                                                                        powershell
PS C:\Users\tharat\OneDrive\เดสก์ท็อป\NetWork\week01-mime-typing> python client/m
ain_enhanced.py assets/sample.png --host 192.168.20.20 --port 65432
[23:53:17] INFO: Starting client with 1 files to 192.168.20.20:65432
[23:53:27] ERROR: Failed to connect to 192.168.20.20:65432: timed out
[23:53:27] ERROR: Some files failed to send
PS C:\Users\tharat\OneDrive\เดสก์ท็อป\NetWork\week01-mime-typing> python client/m
ain_enhanced.py assets/sample.png --host 127.0.0.1 --port 65432
[23:54:46] INFO: Starting client with 1 files to 127.0.0.1:65432
[23:54:46] INFO: Connected to 127.0.0.1:65432
[23:54:46] INFO: Sent assets/sample.png (108 bytes) as image/png
[23:54:46] INFO: Connection closed. Sent: 1, Failed: 0
[23:54:46] INFO: All files sent successfully
PS C:\Users\tharat\OneDrive\เดสก์ท็อป\NetWork\week01-mime-typing>
```

Confirm file saved.

📡 Part 4 — Wireshark Capture Walkthrough on PC1 and PC1

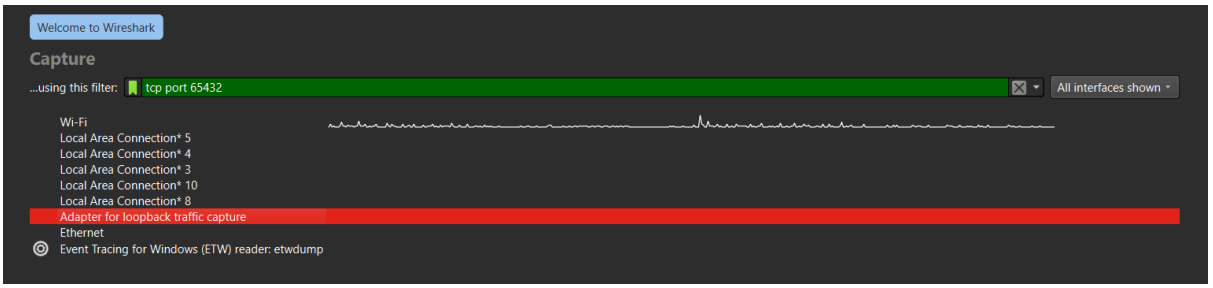
Step 1 — Start Capture

Open Wireshark.

Select interface connected to network.

Apply capture filter:

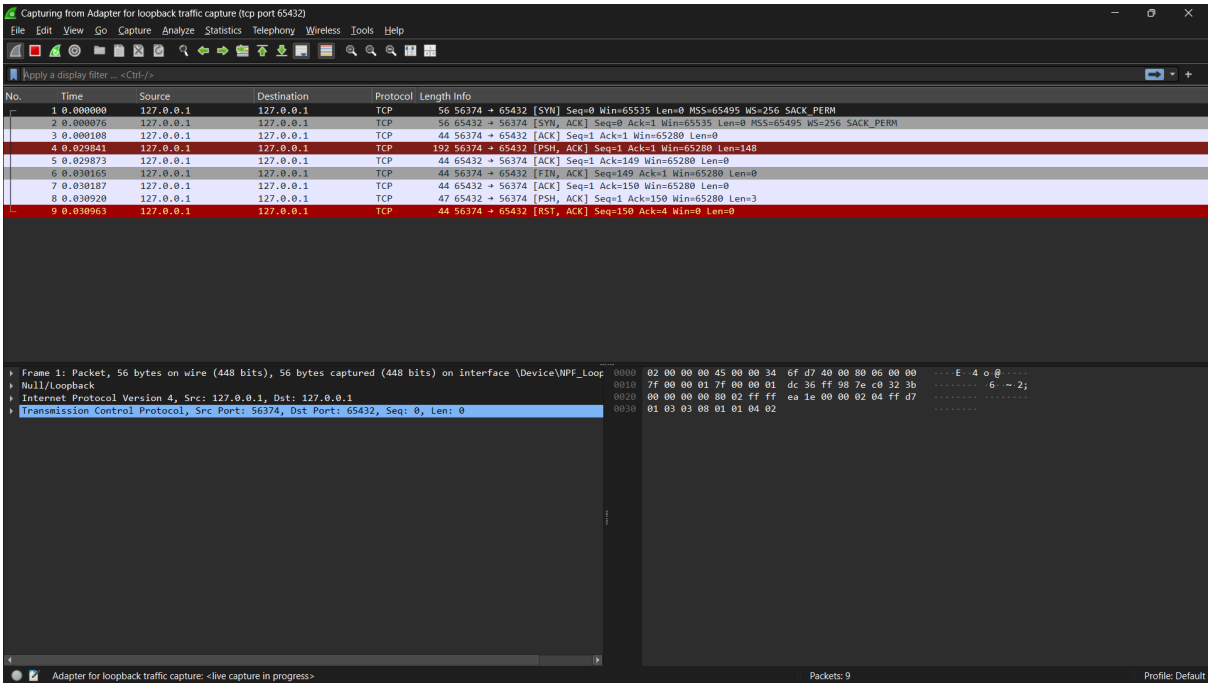
```
None
tcp port 65432
```



Step 2 — Initiate Transfer

Send file again.

Stop capture after completion.



Step 3 — Analyze Packets

A — ARP Exchange

Look for:

- Who has 192.168.20.20?
  ไม่มี ARP เพราะมันไม่ออก LAN

Confirm MAC resolution.

B — TCP Three-Way Handshake

Identify:

1 SYN 2 SYN-ACK 3 ACK

ขั้นตอน TCP 3-way handshake เกิดขึ้นในแพ็กเก็ตที่ 1–3 ดังนี้:

- แพ็กเก็ตที่ 1: Client ส่ง **SYN** ไปยังพอร์ต 65432
- แพ็กเก็ตที่ 2: Server ตอบกลับด้วย **SYN, ACK**
- แพ็กเก็ตที่ 3: Client ส่ง **ACK** กลับไป

กระบวนการนี้ทำให้การเชื่อมต่อ TCP สำเร็จ

Check sequence numbers.

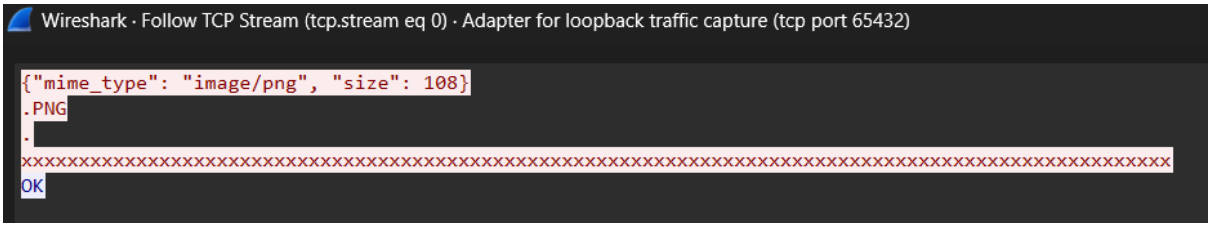
C — Data Transfer

Follow stream:

Right click → Follow → TCP Stream.

You should see JSON header similar to:

```
None
{"mime_type":"image/png","size":12345}
```



Followed by binary.

D — VLAN Tags

If capturing on trunk:

Look for:

```
None
802.1Q Virtual LAN
```

Note VLAN ID.

🔬 Packet Dissection Guide

Layer	Field	What to Observe
Ethernet	Src/Dst MAC	Frame direction
802.1Q	VLAN ID	Segmentation

Layer	Field	What to Observe
IP	TTL	Routing behavior
TCP	Flags	Reliability
Payload	JSON	MIME metadata

🛠️ Part 5 — Troubleshooting Labs

Scenario 1 — Broken Routing

Shutdown subinterface.

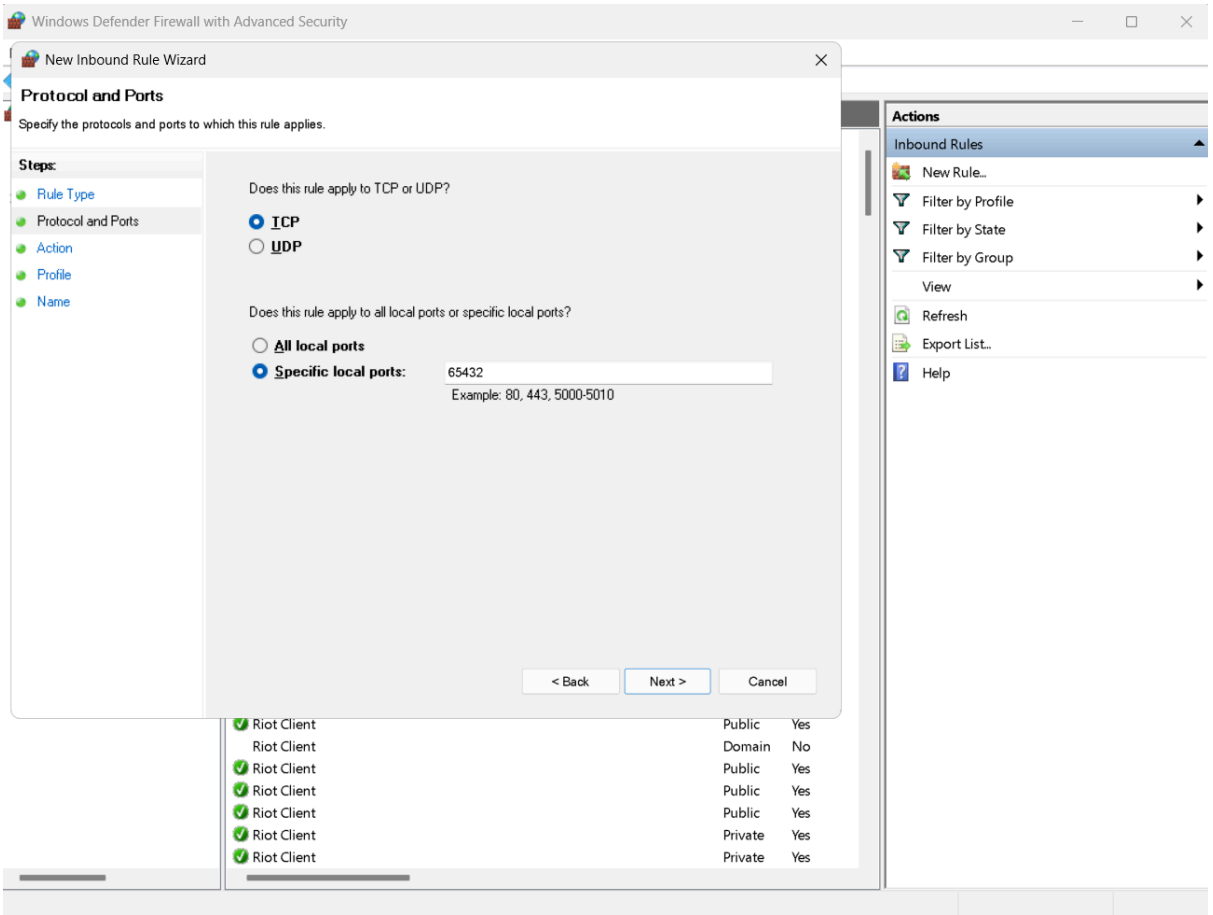
```
PS C:\Users\tharat\OneDrive\เดสก์ท็อป\NetWork\week01-mime-typing> python client/main_
enhanced.py assets/sample.png --host 127.0.0.1 --port 9999
[00:29:15] INFO: Starting client with 1 files to 127.0.0.1:9999
[00:29:17] ERROR: Failed to connect to 127.0.0.1:9999: [WinError 10061] No connecti
on could be made because the target machine actively refused it
[00:29:17] ERROR: Some files failed to send
PS C:\Users\tharat\OneDrive\เดสก์ท็อป\NetWork\week01-mime-typing>
```

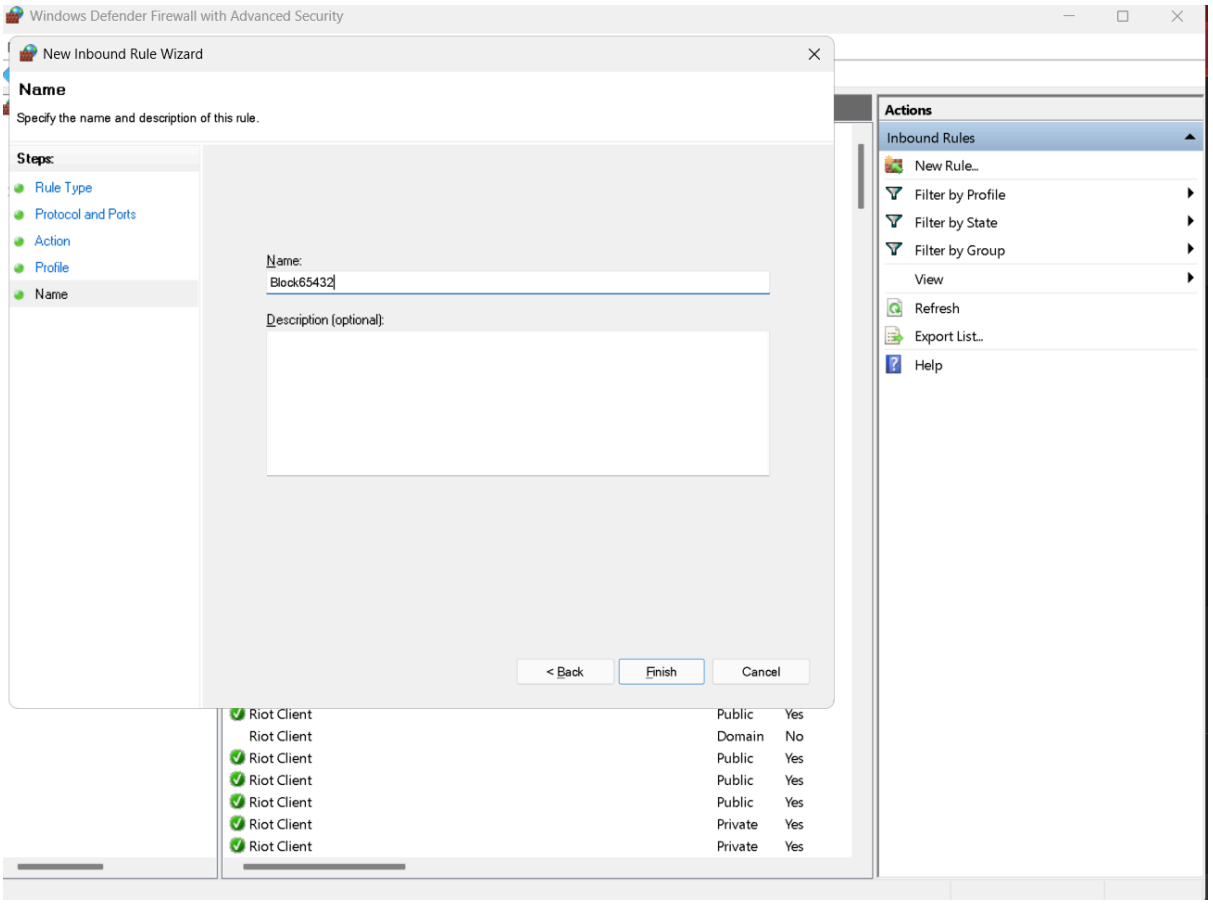
Observe:

- Ping failure
- TCP timeout

Scenario 2 — Firewall Block

Block port.


Windows Defender Firewall with Advanced Security
New Inbound Rule Wizard
Protocol and Ports
Specify the protocols and ports to which this rule applies.
Steps:
Rule Type
Protocol and Ports
Action
Profile
Name
Does this rule apply to TCP or UDP?
TCP
UDP
Does this rule apply to all local ports or specific local ports?
All local ports
Specific local ports:
65432
Example: 80, 443, 5000-5010
< Back
Next >
Cancel
Actions
Inbound Rules
New Rule...
Filter by Profile
Filter by State
Filter by Group
View
Refresh
Export List...
Help
Riot Client Public Yes
Riot Client Domain No
Riot Client Public Yes
Riot Client Public Yes
Riot Client Public Yes
Riot Client Private Yes
Riot Client Private Yes


Windows Defender Firewall with Advanced Security
New Inbound Rule Wizard
Name
Specify the name and description of this rule.
Steps:
Rule Type
Protocol and Ports
Action
Profile
Name
Name:
Block65432
Description (optional):
< Back
Finish
Cancel
Actions
Inbound Rules
New Rule...
Filter by Profile
Filter by State
Filter by Group
View
Refresh
Export List...
Help
Riot Client Public Yes
Riot Client Domain No
Riot Client Public Yes
Riot Client Public Yes
Riot Client Public Yes
Riot Client Private Yes
Riot Client Private Yes

127.0.0.1 (loopback)

Windows Firewall ปกติ ไม่ block loopback traffic

เพราะมันคือการคุยภายในเครื่องตัวเอง ดังนั้น block port แล้วก็ยังวิ่งได้เหมือนเดิม

Observe:

- SYN retransmissions

Scenario 3 — Wrong Gateway

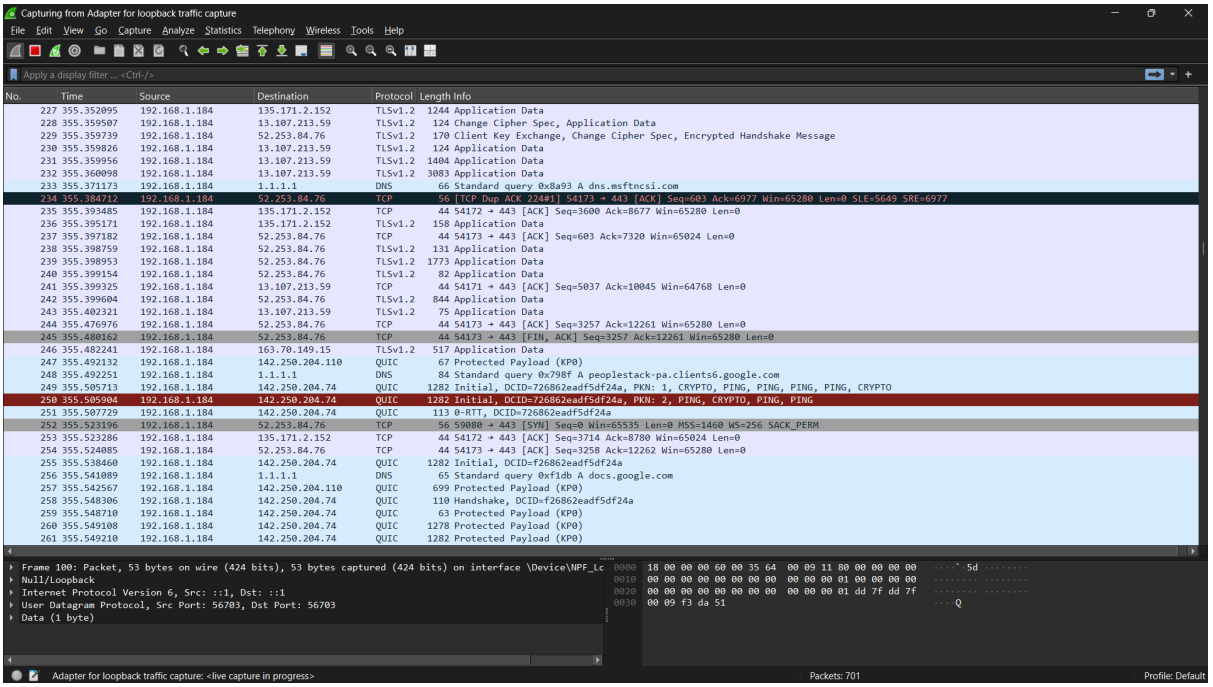
```
Pinging 192.168.20.20 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.20:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PS C:\Users\tharat>
```

Observe:

- ARP incomplete

Scenario 4 — Packet Loss Simulation

Drop packets.



Observe:

- TCP retransmissions
- Window adjustment

📝 Lab Report Requirements

Include:

- Screenshots of configurations
- Ping results
- Wireshark capture screenshots
- JSON header evidence
- Encapsulation explanation
- Troubleshooting notes

🎓 Assessment Questions

1. Why does TCP retransmit lost segments?

   TCP ส่งข้อมูลซ้ำเมื่อเกิดการสูญหาย (Retransmit) เพราะ TCP เป็นโปรโตคอลที่ต้องการความน่าเชื่อถือในการรับส่งข้อมูลเมื่อเกิดการสูญหายของข้อมูล ผู้รับจะส่ง Duplicate ACK กลับมา หรือผู้ส่งตรวจพบ Timeout จากนั้น TCP จะส่งข้อมูลส่วนนั้นซ้ำอีกครั้ง เพื่อให้มั่นใจว่าข้อมูลครบถ้วนและเรียงลำดับถูกต้อง

2. How does VLAN tagging affect broadcast domains?

VLAN Tagging ทำให้เครือข่ายถูกแบ่งออกเป็น Broadcast Domain แยกจากกันในเชิงตรรกะอุปกรณ์ที่อยู่คนละ VLAN จะไม่สามารถรับ Broadcast จากกันได้ เว้นแต่มีการทำ Routing ระหว่าง VLANการใช้ VLAN ช่วยเพิ่มความปลอดภัย ลดปริมาณ Broadcast และควบคุมทราฟฟิกในเครือข่ายได้ดีขึ้น

3. What role does MIME play compared to TCP headers?
   MIME มีหน้าที่กำหนดประเภทและความหมายของข้อมูลในระดับ Application เช่น image/png หรือ text/plain ส่วน TCP Header มีหน้าที่ควบคุมการรับส่งข้อมูลให้ถูกต้อง เช่น การจัดลำดับข้อมูล การยืนยันการรับข้อมูล และการควบคุมการไหลของข้อมูล

4. Where does routing decision occur?
   การตัดสินใจเลือกเส้นทาง (Routing Decision) เกิดขึ้นที่ชั้น Network Layer (Layer 3) Router จะตรวจสอบค่า Destination IP Address ใน IP Header และเปรียบเทียบกับ Routing Table เพื่อเลือกเส้นทางถัดไป

5. Why does ARP precede TCP communication?
   ARP เกิดก่อน TCP เพราะในเครือข่าย LAN จำเป็นต้องทราบ MAC Address ของปลายทางก่อนจึงจะสามารถส่งข้อมูลผ่าน Ethernet ได้แม้ TCP จะทำงานที่ Layer 4 แต่การส่งข้อมูลจริงในเครือข่ายท้องถิ่นต้องใช้ MAC Address ซึ่งได้มาจากกระบวนการ ARPดังนั้น ARP จึงเกิดขึ้นก่อนการสื่อสาร TCP เสมอใน LAN

📊 Reflection Exercise

Explain:

How does application meaning survive multiple encapsulation layers?
ความหมายของข้อมูลระดับ Application ยังคงอยู่แม้ผ่านหลายชั้นของการห่อหุ้ม (Encapsulation) เพราะแต่ละ Layer จะเพิ่ม Header ของตนเองเข้าไปโดยไม่แก้ไขข้อมูลภายในเมื่อข้อมูลถึงปลายทาง จะมีการ Decapsulation ทีละชั้น โดยลบ Header ออกตามลำดับ จนเหลือข้อมูล Application เดิมที่ไม่ถูกเปลี่ยนแปลง

⚠️ Common Pitfalls

- Server bound to localhost only
- Missing trunk
- Wrong VLAN
- OS firewall enabled
- Capture on wrong interface

🚀 Extension Activities

- Add TLS encryption
- Measure throughput vs latency
- Capture checksum errors
- Add checksum field to protocol
- Containerize server
- Compare with HTTP transfer

🧬 Instructor Notes — What This Lab Really Teaches

This is not just networking.

It demonstrates:

- Protocol layering philosophy
- Reliability engineering
- Metadata semantics
- Distributed system fundamentals

The student watches a file become:

intention → bytes → frames → signals → bytes → meaning

That transformation is the essence of computer networks.

นายเกียรติศักดิ์ นันทรัตน์ 673380072-9 รับผิดชอบ Part 1,2,3
นายสรวิชญ์ วันเสน 673380064-8 รับผิดชอบ Part 4,5
นายศุภวัฒน์ ข่ายทอง 673380062-2 รับผิดชอบ Assessment Questions,Reflection Exercise